NEWTON SCHOOL

PRE-CALCULUS | U4 PBA

Newton's Law of Cooling — Hot Coffee PBA

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Topic: *Exponential decay, horizontal asymptotes, Newton's Law of Cooling. Unit 4 PBA reference + in-class worked problems.*

☕ How to use this note

Read the **Notes** section (the big formula + every variable) carefully first. Then go through all three **Worked Examples** — follow every step. Finally, attempt each **Practice Problem** in the dashed work-space *before* reading the green solution box. The boxed green answer is the final answer to check yourself.

📘 Notes — Newton's Law of Cooling Toolkit

1. The Main Formula

A hot (or cold) object placed in a room of constant temperature T_{room} follows:

$$T(t) \;=\; T_{\text{room}} \;+\; (T_0 - T_{\text{room}})\, e^{-kt}$$

- $T(t)$ = temperature at time t (in °F or °C)
- T_0 = **initial temperature** (at $t = 0$)
- T_{room} = **room (ambient) temperature** — this is the *horizontal asymptote*
- k = **cooling rate constant** ($k > 0$; bigger k = cools faster)
- t = time (minutes or hours — be consistent!)

2. The Horizontal Asymptote — Why It Matters

As $t \to \infty$, the term $e^{-kt} \to 0$, so:

$$\lim_{t\to\infty} T(t) \;=\; T_{\text{room}} \;+\; (T_0 - T_{\text{room}}) \cdot 0 \;=\; T_{\text{room}}$$

The coffee can never get colder than room temperature. The graph starts at T_0, curves down (for hot coffee), and approaches T_{room} — never touching it. On your PBA: the horizontal asymptote of the graph *is* the room temperature.

3. Solving for Time — "When does it reach T_1?"

Set $T(t) = T_1$ and solve for t:

$$T_1 = T_{\text{room}} + (T_0 - T_{\text{room}})\, e^{-kt}$$

$$e^{-kt} = \frac{T_1 - T_{\text{room}}}{T_0 - T_{\text{room}}}$$

$$-kt = \ln\!\left(\frac{T_1 - T_{\text{room}}}{T_0 - T_{\text{room}}}\right)$$

$$\boxed{t \;=\; \frac{1}{k}\ln\!\left(\frac{T_0 - T_{\text{room}}}{T_1 - T_{\text{room}}}\right)}$$

(Flip the fraction inside to remove the negative sign — valid since $T_0 > T_1 > T_{\text{room}}$.)

4. Half-Temperature-Difference Time $t_{1/2}$

The time it takes for the *difference* $(T - T_{\text{room}})$ to be cut in half:

$$t_{1/2} \;=\; \frac{\ln 2}{k} \;\approx\; \frac{0.693}{k}$$

Analogy to radioactive half-life — same math, different context.

5. Finding k from Two Data Points

If you measure temperatures T_1 at t_1 and T_2 at t_2:

$$e^{-kt_1} = \frac{T_1 - T_{\text{room}}}{T_0 - T_{\text{room}}}, \quad e^{-kt_2} = \frac{T_2 - T_{\text{room}}}{T_0 - T_{\text{room}}}$$

$$\text{Divide:} \quad e^{-k(t_2 - t_1)} = \frac{T_2 - T_{\text{room}}}{T_1 - T_{\text{room}}}$$

$$k = \frac{1}{t_1 - t_2} \ln\left(\frac{T_2 - T_{\text{room}}}{T_1 - T_{\text{room}}}\right) = \frac{1}{t_2 - t_1} \ln\left(\frac{T_1 - T_{\text{room}}}{T_2 - T_{\text{room}}}\right)$$

Typical values for context:

Room temp: $T_{\text{room}} \approx 68\text{–}72\ ^\circ\text{F}$ · Fresh-brewed coffee: $T_0 \approx 175\text{–}185\ ^\circ\text{F}$ · Drinkable: $T_1 \approx 130\text{–}140\ ^\circ\text{F}$

Ceramic mug: $k \approx 0.010\text{–}0.015/\text{min}$ · Paper cup: $k \approx 0.04\text{–}0.06/\text{min}$

Why is the horizontal asymptote exactly T_{room}?

The exponential factor e^{-kt} approaches **zero** as $t \to \infty$ (because the exponent $-kt \to -\infty$). So the whole $(T_0 - T_{\text{room}})\, e^{-kt}$ piece vanishes — what's left is just T_{room}. Geometrically: the graph is the parent curve e^{-kt} scaled and shifted *up* by T_{room}. The "floor" moves from 0 to T_{room}. That shift *is* the asymptote.

Worked Examples

Example 1 — Predict temperature at a given time

> A freshly poured mug of coffee starts at $T_0 = 180\ ^\circ\text{F}$. The room temperature is $T_{\text{room}} = 72\ ^\circ\text{F}$. The cooling constant is $k = 0.025\ \text{min}^{-1}$. What is the temperature after 30 minutes?

Setup. We know $T_0 = 180, T_{\text{room}} = 72, k = 0.025, t = 30$.

Plug in. $T(30) = 72 + (180 - 72)\, e^{-0.025 \times 30}$

$$= 72 + 108\, e^{-0.75}$$
$$= 72 + 108 \times 0.4724$$
$$= 72 + 51.02 \approx 123.0\ ^\circ\text{F}.$$

Answer. $\boxed{T(30) \approx 123\ ^\circ\text{F}}$

Sanity check. After 30 min the coffee went from 180 °F down to ~123 °F — cooled by 57 °F. Still hotter than room temp (72 °F). Makes sense.

Example 2 — Find k from two measurements

You measure a cup of coffee at $t = 0$ min: $T_0 = 175\ ^\circ\mathrm{F}$. At $t = 10$ min: $T = 155\ ^\circ\mathrm{F}$. At $t = 20$ min: $T = 138\ ^\circ\mathrm{F}$. Room temperature is $T_{\text{room}} = 70\ ^\circ\mathrm{F}$. Find the cooling constant k.

Strategy. Use the formula at $t = 10$:

$$155 = 70 + (175 - 70)\, e^{-k \cdot 10}$$

$$155 - 70 = 105\, e^{-10k} \;\Rightarrow\; 85 = 105\, e^{-10k}$$

$$e^{-10k} = \frac{85}{105} = 0.8095$$

$$-10k = \ln(0.8095) = -0.2113 \;\Rightarrow\; k = 0.02113\,\mathrm{min}^{-1}.$$

Verify at $t = 20$.

$$T(20) = 70 + 105\, e^{-0.02113 \times 20} = 70 + 105 \times e^{-0.4226} = 70 + 105 \times 0.6551 = 70 + 68.8 \approx 138.8\ ^\circ\mathrm{F}.$$

Matches the given $138\ ^\circ\mathrm{F}$.

Answer. $\boxed{k \approx 0.0211\,\mathrm{min}^{-1}}$

Example 3 — Solve for time

Coffee starts at $T_0 = 180\ ^\circ\mathrm{F}$. Room is $72\ ^\circ\mathrm{F}$. The cooling constant $k = 0.025\,\mathrm{min}^{-1}$. How long until the coffee cools to a drinkable $140\ ^\circ\mathrm{F}$?

Set up the equation. $140 = 72 + (180 - 72)\, e^{-0.025t}$

$$140 - 72 = 108\, e^{-0.025t} \;\Rightarrow\; 68 = 108\, e^{-0.025t}$$

$$e^{-0.025t} = \frac{68}{108} = 0.6296$$

$$-0.025t = \ln(0.6296) = -0.4626$$

$$t = \frac{0.4626}{0.025} = 18.5\,\mathrm{min}.$$

Or using the closed-form:

$$t = \frac{1}{0.025} \ln\!\left(\frac{180 - 72}{140 - 72}\right) = 40\ \ln\!\left(\frac{108}{68}\right) = 40\ \ln(1.5882) \approx 40 \times 0.4626 = 18.5\,\mathrm{min}.$$

Answer. $\boxed{t \approx 18.5 \text{ minutes}}$

Practice Problems

Problem 1 Temperature after time — ceramic mug

STATEMENT

A ceramic mug of coffee starts at $T_0 = 185\ ^\circ\mathrm{F}$ in a room at $T_{\text{room}} = 68\ ^\circ\mathrm{F}$. The cooling constant for this mug is $k = 0.018\ \mathrm{min}^{-1}$.

1. Write the complete equation $T(t)$ for this situation.
2. What is the temperature after 15 minutes?
3. What is the horizontal asymptote? What does it represent physically?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Equation. $T(t) = 68 + (185 - 68)\, e^{-0.018t} = 68 + 117\, e^{-0.018t}$

$$\boxed{T(t) = 68 + 117e^{-0.018t}}$$

(b) At $t = 15$. $T(15) = 68 + 117\, e^{-0.018\times 15} = 68 + 117\, e^{-0.27}$

$e^{-0.27} \approx 0.7634$

$T(15) = 68 + 117 \times 0.7634 = 68 + 89.3 \approx 157.3\ ^\circ\mathrm{F}$

$$\boxed{T(15) \approx 157\ ^\circ\mathrm{F}}$$

(c) Horizontal asymptote. As $t \to \infty$, $e^{-0.018t} \to 0$, so $T \to 68$.

$$\boxed{\textbf{Horizontal asymptote: } T = 68\ ^\circ\mathrm{F} \textbf{ = room temperature}}$$

Physical meaning: the coffee will never drop below 68 °F (room temp). It asymptotically approaches but never crosses.

Problem 2 Find k from one measurement

STATEMENT

A cup of coffee starts at $T_0 = 180\,^\circ\mathrm{F}$ in a $70\,^\circ\mathrm{F}$ room. After 20 minutes, the coffee is $130\,^\circ\mathrm{F}$.

1. Find the cooling constant k.
2. Write the equation $T(t)$.

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Solve for k. At $t = 20$: $130 = 70 + (180 - 70)\, e^{-20k}$

$60 = 110\, e^{-20k}$

$e^{-20k} = \dfrac{60}{110} = 0.5\overline{45}$

$-20k = \ln(60/110) = \ln(6/11) \approx -0.6061$

$k = \dfrac{0.6061}{20} \approx 0.0303\,\mathrm{min}^{-1}$

$\boxed{k \approx 0.0303\,\mathrm{min}^{-1}}$

(b) Full equation.

$\boxed{T(t) = 70 + 110\, e^{-0.0303t}}$

Problem 3 When does it reach drinking temperature?

STATEMENT

Coffee starts at $180\,^\circ\mathrm{F}$ in a $72\,^\circ\mathrm{F}$ room with $k = 0.030\,\mathrm{min}^{-1}$. You like to drink coffee at exactly $140\,^\circ\mathrm{F}$.

1. How long do you have to wait?
2. Find $t_{1/2}$ (the half-temperature-difference time).

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Solve for t. $140 = 72 + (180 - 72)\,e^{-0.030t}$

$68 = 108\,e^{-0.030t}$

$e^{-0.030t} = \dfrac{68}{108} \approx 0.6296$

$-0.030t = \ln(0.6296) \approx -0.4626$

$t = \dfrac{0.4626}{0.030} \approx 15.4\,\mathrm{min}$

$\boxed{t \approx 15.4\ \textbf{minutes}}$

(b) Half-temperature-difference time. $t_{1/2} = \dfrac{\ln 2}{k} = \dfrac{0.6931}{0.030} \approx 23.1\,\mathrm{min}$

Meaning: at $t = 23.1$ min the temperature difference from room is cut in half.

Check: difference starts at $180 - 72 = 108$ °F. Half = 54 °F. So $T = 72 + 54 = 126\,^\circ\mathrm{F}$.

$\boxed{t_{1/2} \approx 23.1\ \textbf{minutes}}$

Problem 4 Identify room temperature from a graph

STATEMENT

A cooling curve is given by $T(t) = 65 + 120\,e^{-0.022t}$.

1. What is the room temperature (horizontal asymptote)?
2. What was the initial temperature T_0?
3. What is the temperature at $t = 60$ minutes?
4. At what time does the coffee reach $90\ ^{\circ}\mathrm{F}$?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Horizontal asymptote. As $e^{-0.022t} \to 0$: $T \to 65$.

$$\boxed{T_{\text{room}} = 65\ ^{\circ}\mathrm{F}}$$

(b) Initial temperature. $T(0) = 65 + 120\,e^{0} = 65 + 120 = 185$.

$$\boxed{T_0 = 185\ ^{\circ}\mathrm{F}}$$

(c) At $t = 60$. $T(60) = 65 + 120\,e^{-0.022\times 60} = 65 + 120\,e^{-1.32}$

$e^{-1.32} \approx 0.2671$

$T(60) = 65 + 120 \times 0.2671 = 65 + 32.05 \approx 97.1\ ^{\circ}\mathrm{F}$

$$\boxed{T(60) \approx 97\ ^{\circ}\mathrm{F}}$$

(d) When does $T = 90$? $90 = 65 + 120\,e^{-0.022t}$

$25 = 120\,e^{-0.022t} \;\Rightarrow\; e^{-0.022t} = 25/120 = 0.2083$

$-0.022t = \ln(0.2083) \approx -1.5686$

$t \approx 71.3\,\mathrm{min}$

$$\boxed{t \approx 71.3 \textbf{ minutes}}$$

Problem 5 Two-point data fit + prediction

STATEMENT

A scientist measures cooling data in a $68\,^{\circ}\mathrm{F}$ lab:

- At $t = 5$ min: $T = 160\,^{\circ}\mathrm{F}$
- At $t = 15$ min: $T = 130\,^{\circ}\mathrm{F}$

1. Find the initial temperature T_0 and the cooling constant k. *(Hint: set up two equations and divide to eliminate T_0.)*
2. Predict the temperature at $t = 40$ min.
3. When will the object be within $5\,^{\circ}\mathrm{F}$ of room temperature?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Set up equations. Let $A = T_0 - 68$. Then $T(t) = 68 + A\,e^{-kt}$.

At $t = 5$: $160 - 68 = 92 = A\,e^{-5k}$ ...(1)

At $t = 15$: $130 - 68 = 62 = A\,e^{-15k}$ ...(2)

Divide (2) ÷ (1): $\dfrac{62}{92} = \dfrac{A\,e^{-15k}}{A\,e^{-5k}} = e^{-10k}$

$e^{-10k} = 0.6739 \Rightarrow -10k = \ln(0.6739) = -0.3941 \Rightarrow k \approx 0.03941$

Find A from (1): $A = 92\,/\,e^{-5\times 0.03941} = 92\,/\,e^{-0.1971} = 92\,/\,0.8211 \approx 112.0$

So $T_0 = 68 + 112 = 180\ ^\circ\mathrm{F}$.

$\boxed{T_0 \approx 180\ ^\circ\mathrm{F},\ k \approx 0.0394\,\mathrm{min}^{-1}}$

Full equation: $T(t) = 68 + 112\,e^{-0.0394t}$

(b) At $t = 40$. $T(40) = 68 + 112\,e^{-0.0394\times 40} = 68 + 112\,e^{-1.576}$

$e^{-1.576} \approx 0.2068$

$T(40) = 68 + 112 \times 0.2068 \approx 68 + 23.2 \approx 91.2\ ^\circ\mathrm{F}$

$\boxed{T(40) \approx 91\ ^\circ\mathrm{F}}$

(c) Within 5 °F of room ($T = 73\ ^\circ\mathrm{F}$). $73 = 68 + 112\,e^{-0.0394t} \Rightarrow 5 = 112\,e^{-0.0394t}$

$e^{-0.0394t} = 5/112 = 0.04464$

$-0.0394t = \ln(0.04464) = -3.109 \Rightarrow t \approx 78.9\,\mathrm{min}$

$\boxed{t \approx 79\ \text{minutes}}$

Problem 6 Warming coffee — a reverse scenario

STATEMENT

An iced drink starts at $T_0 = 38\ ^\circ\mathrm{F}$ in a $75\ ^\circ\mathrm{F}$ room. The same formula applies: $T(t) = T_{\text{room}} + (T_0 - T_{\text{room}})\,e^{-kt}$, but now $(T_0 - T_{\text{room}}) < 0$, so the drink *warms*. Given $k = 0.035\,\mathrm{min}^{-1}$:

1. What is the horizontal asymptote? What does it represent?
2. After 20 minutes, what is the temperature of the drink?
3. How long until the drink reaches $60\ ^\circ\mathrm{F}$?

WORK SPACE — YOUR SOLUTION

SOLUTION

(a) Horizontal asymptote. As $t \to \infty$: $T \to T_{\text{room}} = 75\ ^\circ\text{F}$. The drink asymptotically warms toward room temperature.

$\boxed{\textbf{Asymptote: } T = 75\ ^\circ\text{F}}$

(b) At $t = 20$. $T(20) = 75 + (38 - 75)\, e^{-0.035\times 20} = 75 + (-37)\, e^{-0.70}$

$e^{-0.70} \approx 0.4966$

$T(20) = 75 - 37 \times 0.4966 = 75 - 18.4 \approx 56.6\ ^\circ\text{F}$

$\boxed{T(20) \approx 56.6\ ^\circ\text{F}}$

(c) When $T = 60\ ^\circ\text{F}$. $60 = 75 + (-37)\, e^{-0.035t} \;\Rightarrow\; -15 = -37\, e^{-0.035t}$

$e^{-0.035t} = \dfrac{15}{37} = 0.4054$

$-0.035t = \ln(0.4054) = -0.9027$

$t = \dfrac{0.9027}{0.035} \approx 25.8\,\text{min}$

$\boxed{t \approx 25.8 \textbf{ minutes}}$

Key insight. The same formula works for both cooling AND warming. The sign of $(T_0 - T_{\text{room}})$ determines the direction. The asymptote is always T_{room} — the environment always wins.